

PROGETTO S9/L5

Traccia:

Per l'esercizio pratico di oggi, trovate in allegato una cattura di rete effettuata con Wireshark. Analizzate la cattura attentamente e rispondere ai seguenti quesiti:

- Identificare ed analizzare eventuali IOC, ovvero evidenze di attacchi in corso;
- In base agli IOC trovati, fate delle ipotesi sui potenziali vettori di attacco utilizzati;
- Consigliate un'azione per ridurre gli impatti dell'attacco attuale ed eventualmente un simile attacco futuro.

Svolgimento dell'esercizio

Dopo aver scaricato l'allegato, ho creato una cartella condivisa dal mio pc alla vm kali per poter passare il file sulla vm.

Dopo averlo spostato sul desktop e aperto con Wireshark, si vede che il file `"Cattura_U3_W1_L3.pcapng"` contiene il traffico di rete, registrato sempre con Wireshark, tra due computer sulla stessa rete locale:

- 192.168.200.100 è il computer che invia il traffico;
- 192.168.200.150 è il computer che riceve il traffico

File
Edit
View
Go
Capture
Analyze
Statistics
Telephony
Wireless
Tools
Help

Time display filter on <Ctrl>F

No.	Time	Source	Destination	Protocol	Length	Info
1	1.0000000	192.168.200.150	192.168.200.250	OSPF	28	OSPF Hello
2	63.76421495	192.168.200.150	192.168.200.150	TCP	74	53960 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810522428 TSecr=0 WS=128
3	63.76427879	192.168.200.150	192.168.200.150	TCP	74	53876 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810522428 TSecr=0 WS=128
4	63.76417732	192.168.200.150	192.168.200.150	TCP	74	80 -> 53060 [SYN] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945116 TSecr=810522427 WS=64
5	63.76417732	192.168.200.150	192.168.200.150	TCP	74	53876 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
6	63.764851289	192.168.200.150	192.168.200.150	TCP	66	53608 -> 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810522428 TSecr=424945116
7	63.764851289	192.168.200.150	192.168.200.150	TCP	66	53608 -> 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810522428 TSecr=424945116
8	63.76492461	PCSSystemtic.fid.871...	PCSSystemtic.fid.871...	TCP	60	Who has 192.168.200.100? Ttl=192.168.200.150
9	63.76492461	PCSSystemtic.fid.871...	PCSSystemtic.fid.871...	TCP	42	192.168.200.100 is at 00:00:27:39:7d:fe
10	63.764852257	PCSSystemtic.fid.871...	PCSSystemtic.fid.871...	TCP	42	Who has 192.168.200.150? Ttl=192.168.200.150
11	63.76430099	192.168.200.150	192.168.200.150	TCP	74	53960 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535437 TSecr=0 WS=128
12	63.76431445	192.168.200.150	192.168.200.150	TCP	74	53876 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535437 TSecr=0 WS=128
13	63.76436305	192.168.200.150	192.168.200.150	TCP	74	53876 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
14	63.76436305	192.168.200.150	192.168.200.150	TCP	74	53876 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
15	63.76436305	192.168.200.150	192.168.200.150	TCP	74	53876 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
16	63.76436305	192.168.200.150	192.168.200.150	TCP	74	53876 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
17	63.76436305	192.168.200.150	192.168.200.150	TCP	74	53876 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
18	63.76436305	192.168.200.150	192.168.200.150	TCP	74	53876 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
19	63.76436305	192.168.200.150	192.168.200.150	TCP	74	53876 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
20	63.76436305	192.168.200.150	192.168.200.150	TCP	74	53876 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
21	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
22	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
23	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
24	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
25	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
26	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
27	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
28	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
29	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
30	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
31	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
32	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
33	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
34	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
35	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
36	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
37	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
38	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
39	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
40	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
41	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
42	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
43	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
44	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
45	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
46	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
47	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
48	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
49	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
50	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
51	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
52	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
53	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
54	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
55	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
56	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
57	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
58	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
59	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
60	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
61	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
62	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
63	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
64	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
65	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
66	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
67	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
68	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
69	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
70	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
71	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
72	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
73	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
74	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
75	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
76	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
77	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
78	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
79	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
80	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 -> 80 [ACK] Seq=1 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=424945246 TSecr=810535437 WS=64
81	63.76485695	192.168.200.150	192.168.200.150	TCP	66	53428 ->

Dalla prima riga si nota subito che il computer 192.168.200.150 comunica in broadcast il proprio nome "METASPLOITABLE".

Subito dopo il computer 192.168.200.100 inizia a mandare pacchetti SYN verso il computer 192.168.200.150, provando ad aprire una connessione verso tante porte diverse iniziando da quelle più famose.

Per ogni tentativo il bersaglio risponde in due modi possibili

- RST, ACK significa che la porta è chiusa e non c'è nessun servizio;
- SYN, ACK significa che la porta è aperta e c'è un servizio in ascolto

Questo comportamento indica che è in atto una scansione delle porte tramite Nmap.

Ho verificato quanto riportato sopra, applicando dei filtri su Wireshark, in particolare “*tcp.flags.syn==1 and tcp.flags.ack==0*” per filtrare solo i tentativi di connessione in uscita (SYN), che mostra tutte le porte che 192.168.200.100 ha provato a raggiungere. Dopodichè ho applicato il filtro “*tcp.flags.reset==1*” che indica tutte le risposte di rifiuto (RST, ACK), vedendo che quasi tutti i tentativi risultano negativi e che quindi si tratta di una scansione a tappeto e non di traffico normale di rete.

Cattura_U3_W1_L5.pcapng						
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
tcp.flags.syn==1 and tcp.flags.ack==0						
No.	Time	Source	Destination	Protocol	Length	Info
2	23.764214995	192.168.200.100	192.168.200.150	TCP	74	53060 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810522427 TSecr=0 WS=128
3	23.764267789	192.168.200.100	192.168.200.150	TCP	74	33876 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810522428 TSecr=0 WS=128
12	36.774143445	192.168.200.100	192.168.200.150	TCP	74	41304 → 23 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535437 TSecr=0 WS=128
13	36.774218116	192.168.200.100	192.168.200.150	TCP	74	56120 → 111 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535437 TSecr=0 WS=128
14	36.774257841	192.168.200.100	192.168.200.150	TCP	74	33878 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535437 TSecr=0 WS=128
15	36.774366385	192.168.200.100	192.168.200.150	TCP	74	58636 → 554 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
16	36.774405627	192.168.200.100	192.168.200.150	TCP	74	52358 → 135 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
17	36.774535534	192.168.200.100	192.168.200.150	TCP	74	45138 → 993 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
18	36.774614776	192.168.200.100	192.168.200.150	TCP	74	41182 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
29	36.775337800	192.168.200.100	192.168.200.150	TCP	74	59174 → 113 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535438 TSecr=0 WS=128
30	36.775386694	192.168.200.100	192.168.200.150	TCP	74	55656 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535439 TSecr=0 WS=128
31	36.775524204	192.168.200.100	192.168.200.150	TCP	74	53062 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535439 TSecr=0 WS=128
42	36.776179538	192.168.200.100	192.168.200.150	TCP	74	50684 → 199 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535439 TSecr=0 WS=128
43	36.776233880	192.168.200.100	192.168.200.150	TCP	74	54220 → 985 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535439 TSecr=0 WS=128
44	36.776336810	192.168.200.100	192.168.200.150	TCP	74	34648 → 587 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
45	36.776385694	192.168.200.100	192.168.200.150	TCP	74	33842 → 445 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
46	36.776402500	192.168.200.100	192.168.200.150	TCP	74	49814 → 256 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
49	36.776478201	192.168.200.100	192.168.200.150	TCP	74	46990 → 139 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
59	36.776496366	192.168.200.100	192.168.200.150	TCP	74	53206 → 143 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
51	36.776512221	192.168.200.100	192.168.200.150	TCP	74	60632 → 25 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
52	36.776568606	192.168.200.100	192.168.200.150	TCP	74	49654 → 110 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
53	36.776671271	192.168.200.100	192.168.200.150	TCP	74	37282 → 53 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
54	36.776720715	192.168.200.100	192.168.200.150	TCP	74	54898 → 500 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
56	36.776843423	192.168.200.100	192.168.200.150	TCP	74	51534 → 487 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
79	36.777143014	192.168.200.100	192.168.200.150	TCP	74	56990 → 707 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
71	36.777186821	192.168.200.100	192.168.200.150	TCP	74	35638 → 436 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535440 TSecr=0 WS=128
72	36.777302991	192.168.200.100	192.168.200.150	TCP	74	34120 → 98 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535441 TSecr=0 WS=128
73	36.777337934	192.168.200.100	192.168.200.150	TCP	74	49780 → 78 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535441 TSecr=0 WS=128
76	36.777473918	192.168.200.100	192.168.200.150	TCP	74	36138 → 580 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535441 TSecr=0 WS=128
77	36.777522494	192.168.200.100	192.168.200.150	TCP	74	52428 → 982 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535441 TSecr=0 WS=128
80	36.777645927	192.168.200.100	192.168.200.150	TCP	74	41874 → 764 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535441 TSecr=0 WS=128
81	36.777680898	192.168.200.100	192.168.200.150	TCP	74	51506 → 435 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535441 TSecr=0 WS=128
90	36.778179978	192.168.200.100	192.168.200.150	TCP	74	51450 → 148 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535441 TSecr=0 WS=128
91	36.778200161	192.168.200.100	192.168.200.150	TCP	74	48448 → 806 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535441 TSecr=0 WS=128
92	36.778307630	192.168.200.100	192.168.200.150	TCP	74	54566 → 221 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535442 TSecr=0 WS=128
96	36.778402791	192.168.200.100	192.168.200.150	TCP	74	42420 → 1007 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535442 TSecr=0 WS=128
97	36.778591226	192.168.200.100	192.168.200.150	TCP	74	34646 → 206 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535442 TSecr=0 WS=128
98	36.778614995	192.168.200.100	192.168.200.150	TCP	74	54202 → 131 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535442 TSecr=0 WS=128
101	36.778759636	192.168.200.100	192.168.200.150	TCP	74	40318 → 392 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535442 TSecr=0 WS=128
102	36.778781327	192.168.200.100	192.168.200.150	TCP	74	51276 → 677 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535442 TSecr=0 WS=128
104	36.778864493	192.168.200.100	192.168.200.150	TCP	74	39566 → 856 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535442 TSecr=0 WS=128
107	36.778983153	192.168.200.100	192.168.200.150	TCP	74	47238 → 84 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535442 TSecr=0 WS=128
109	36.779055243	192.168.200.100	192.168.200.150	TCP	74	56642 → 807 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=810535442 TSecr=0 WS=128

tcp.flags.reset==1						
No.	Time	Source	Destination	Protocol	Length	Info
5	23.764777427	192.168.200.150	192.168.200.100	TCP	60	443 → 33876 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
7	23.764899091	192.168.200.100	192.168.200.150	TCP	60	53060 → 80 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810522428 TSecr=4294951165
21	36.774885696	192.168.200.150	192.168.200.100	TCP	60	443 → 33878 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
22	36.774885737	192.168.200.150	192.168.200.100	TCP	60	53060 → 443 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
23	36.774885716	192.168.200.150	192.168.200.100	TCP	60	135 → 52358 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
26	36.775141184	192.168.200.150	192.168.200.100	TCP	60	993 → 45138 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
32	36.775589806	192.168.200.150	192.168.200.100	TCP	60	113 → 59174 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
33	36.775619454	192.168.200.100	192.168.200.150	TCP	60	41304 → 23 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535439 TSecr=4294952466
34	36.775652497	192.168.200.100	192.168.200.150	TCP	60	56120 → 111 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535439 TSecr=4294952466
39	36.775801904	192.168.200.100	192.168.200.150	TCP	60	41182 → 21 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535439 TSecr=4294952466
40	36.775975876	192.168.200.100	192.168.200.150	TCP	60	55656 → 22 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535439 TSecr=4294952466
41	36.776095853	192.168.200.100	192.168.200.150	TCP	60	53062 → 80 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535439 TSecr=4294952466
47	36.776451284	192.168.200.150	192.168.200.100	TCP	60	199 → 50684 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
48	36.776451357	192.168.200.150	192.168.200.100	TCP	60	995 → 54220 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
55	36.776813123	192.168.200.150	192.168.200.100	TCP	60	587 → 34648 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
58	36.776904922	192.168.200.150	192.168.200.100	TCP	60	256 → 49814 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
60	36.776905804	192.168.200.150	192.168.200.100	TCP	60	143 → 33206 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
62	36.776905082	192.168.200.150	192.168.200.100	TCP	60	110 → 49654 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
64	36.776905162	192.168.200.150	192.168.200.100	TCP	60	500 → 54898 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
69	36.777184811	192.168.200.150	192.168.200.100	TCP	60	487 → 51534 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
74	36.777436032	192.168.200.150	192.168.200.100	TCP	60	707 → 56990 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
75	36.777439741	192.168.200.150	192.168.200.100	TCP	60	455 → 35638 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
78	36.777623082	192.168.200.150	192.168.200.100	TCP	60	98 → 34120 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
79	36.777623149	192.168.200.150	192.168.200.100	TCP	60	78 → 49780 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
82	36.777758636	192.168.200.150	192.168.200.100	TCP	60	580 → 36138 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
83	36.777758696	192.168.200.150	192.168.200.100	TCP	60	962 → 52428 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
84	36.777871245	192.168.200.150	192.168.200.100	TCP	60	764 → 41874 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
85	36.777871290	192.168.200.150	192.168.200.100	TCP	60	435 → 51506 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
86	36.777893298	192.168.200.100	192.168.200.150	TCP	60	33042 → 445 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535441 TSecr=4294952466
87	36.777912171	192.168.200.100	192.168.200.150	TCP	60	46990 → 139 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535441 TSecr=4294952466
88	36.777986759	192.168.200.100	192.168.200.150	TCP	60	60632 → 25 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535441 TSecr=4294952466
89	36.778831265	192.168.200.100	192.168.200.150	TCP	60	37282 → 53 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=810535441 TSecr=4294952466
93	36.778305846	192.168.200.150	192.168.200.100	TCP	60	148 → 51450 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
94	36.778385948	192.168.200.150	192.168.200.100	TCP	60	806 → 48448 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
95	36.778449494	192.168.200.150	192.168.200.100	TCP	60	221 → 54566 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
99	36.778603064	192.168.200.150	192.168.200.100	TCP	60	1007 → 42420 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
100	36.778721080	192.168.200.150	192.168.200.100	TCP	60	206 → 34646 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
103	36.778826294	192.168.200.150	192.168.200.100	TCP	60	131 → 54202 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
105	36.778939327	192.168.200.150	192.168.200.100	TCP	60	392 → 40318 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
106	36.778939427	192.168.200.150	192.168.200.100	TCP	60	677 → 51276 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
108	36.779029210	192.168.200.150	192.168.200.100	TCP	60	856 → 39566 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
110	36.779122299	192.168.200.150	192.168.200.100	TCP	60	84 → 47238 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
112	36.779155984	192.168.200.150	192.168.200.100	TCP	60	807 → 56642 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

In più, sempre su Wireshark, andando su “Statistics” e poi su “Conversations” possiamo notare come le porte scansionate sono le prime 1024.

Wireshark - Conversations - Cattura_U3_W1_L5.pcapng														
Apply a display filter ... <Ctrl-/>														
Wireshark - Conversations - Cattura_U3_W1_L5.pcapng														
Conversation Settings														
Name resolution	Address A Port A Address B Port B Packets Bytes Stream ID Packets A → B Bytes A → B Packets B → A Bytes B → A Rel Start Duration Bits/s A → B Bits/s B → A Flows													
Absolute start time	192.168.200.100 59292 192.168.200.150 1024 2 134 bytes 404 1 74 bytes 1 60 bytes 36.823535894 0.000307 0													
Display raw data	192.168.200.100 38352 192.168.200.150 1022 2 134 bytes 594 1 74 bytes 1 60 bytes 36.835362548 0.002629 0													
Limit to display filter	192.168.200.100 32996 192.168.200.150 1021 2 134 bytes 425 1 74 bytes 1 60 bytes 36.819978259 0.000327 0													
	192.168.200.100 33384 192.168.200.150 1020 2 134 bytes 640 1 74 bytes 1 60 bytes 36.83943937 0.000214 0													
	192.168.200.100 40832 192.168.200.150 1019 2 134 bytes 195 1 74 bytes 1 60 bytes 36.796479443 0.000113 0													
	192.168.200.100 57032 192.168.200.150 1018 2 134 bytes 751 1 74 bytes 1 60 bytes 36.849908583 0.000115 0													
	192.168.200.100 36474 192.168.200.150 1017 2 134 bytes 1017 1 74 bytes 1 60 bytes 36.878091593 0.000245 0													
	192.168.200.100 39078 192.168.200.150 1016 2 134 bytes 273 1 74 bytes 1 60 bytes 36.805293988 0.000099 0													
	192.168.200.100 44580 192.168.200.150 1015 2 134 bytes 260 1 74 bytes 1 60 bytes 36.803593326 0.000128 0													
	192.168.200.100 42700 192.168.200.150 1014 2 134 bytes 66 1 74 bytes 1 60 bytes 36.781159927 0.000096 0													
	192.168.200.100 47100 192.168.200.150 1010 2 134 bytes 579 1 74 bytes 1 60 bytes 36.834310416 0.000242 0													
	192.168.200.100 38350 192.168.200.150 1009 2 134 bytes 555 1 74 bytes 1 60 bytes 36.832122250 0.000130 0													
	192.168.200.100 56076 192.168.200.150 1008 2 134 bytes 379 1 74 bytes 1 60 bytes 36.815148201 0.000304 0													
	192.168.200.100 42420 192.168.200.150 1007 2 134 bytes 35 1 74 bytes 1 60 bytes 36.778482791 0.000180 0													
	192.168.200.100 50768 192.168.200.150 1006 2 134 bytes 176 1 74 bytes 1 60 bytes 36.794718336 0.000230 0													
	192.168.200.100 34030 192.168.200.150 1005 2 134 bytes 843 1 74 bytes 1 60 bytes 36.861457795 0.000351 0													
	192.168.200.100 38430 192.168.200.150 1004 2 134 bytes 177 1 74 bytes 1 60 bytes 36.794959325 0.000327 0													
	192.168.200.100 50686 192.168.200.150 1003 2 134 bytes 521 1 74 bytes 1 60 bytes 36.829026937 0.000185 0													
	192.168.200.100 44018 192.168.200.150 1002 2 134 bytes 833 1 74 bytes 1 60 bytes 36.870473797 0.000170 0													
	192.168.200.100 48512 192.168.200.150 1001 2 134 bytes 402 1 74 bytes 1 60 bytes 36.817062050 0.000233 0													
	192.168.200.100 47044 192.168.200.150 1000 2 134 bytes 651 1 74 bytes 1 60 bytes 36.840589537 0.000336 0													
	192.168.200.100 52136 192.168.200.150 999 2 134 bytes 58 1 74 bytes 1 60 bytes 36.780472830 0.000105 0													
	192.168.200.100 42016 192.168.200.150 998 2 134 bytes 775 1 74 bytes 1 60 bytes 36.852255896 0.000157 0													
	192.168.200.100 46014 192.168.200.150 997 2 134 bytes 301 1 74 bytes 1 60 bytes 36.807616422 0.000186 0													
	192.168.200.100 54302 192.168.200.150 996 2 134 bytes 789 1 74 bytes 1 60 bytes 36.853818263 0.000161 0													
	192.168.200.100 54220 192.168.200.150 995 2 134 bytes 13 1 74 bytes 1 60 bytes 36.771233880 0.000117 0													
	192.168.200.100 53502 192.168.200.150 994 2 134 bytes 706 1 74 bytes 1 60 bytes 36.845839112 0.000162 0													
	192.168.200.100 46138 192.168.200.150 993 2 134 bytes 7 1 74 bytes 1 60 bytes 36.774535534 0.000606 0													
	192.168.200.100 50072 192.168.200.150 992 2 134 bytes 314 1 74 bytes 1 60 bytes 36.808922874 0.000462 0													
	192.168.200.100 47332 192.168.200.150 991 2 134 bytes 202 1 74 bytes 1 60 bytes 36.797266748 0.000135 0													
	192.168.200.100 35934 192.168.200.150 990 2 134 bytes 818 1 74 bytes 1 60 bytes 36.856562694 0.000121 0													
	192.168.200.100 60578 192.168.200.150 989 2 134 bytes 802 1 74 bytes 1 60 bytes 36.854971481 0.000285 0													
	192.168.200.100 52470 192.168.200.150 988 2 134 bytes 391 1 74 bytes 1 60 bytes 36.816045918 0.000201 0													
	192.168.200.100 40816 192.168.200.150 987 2 134 bytes 383 1 74 bytes 1 60 bytes 36.815525420 0.000228 0													
	192.168.200.100 40772 192.168.200.150 986 2 134 bytes 890 1 74 bytes 1 60 bytes 36.866250342 0.000150 0													
	192.168.200.100 52668 192.168.200.150 985 2 134 bytes 886 1 74 bytes 1 60 bytes 36.866066571 0.000126 0													
	192.168.200.100 45286 192.168.200.150 984 2 134 bytes 353 1 74 bytes 1 60 bytes 36.812652603 0.000122 0													
	192.168.200.100 35486 192.168.200.150 983 2 134 bytes 495 1 74 bytes 1 60 bytes 36.827054294 0.000114 0													
	192.168.200.100 33280 192.168.200.150 982 2 134 bytes 234 1 74 bytes 1 60 bytes 36.801426683 0.000220 0													
	192.168.200.100 51436 192.168.200.150 981 2 134 bytes 240 1 74 bytes 1 60 bytes 36.801814562 0.000505 0													
	192.168.200.100 45816 192.168.200.150 980 2 134 bytes 245 1 74 bytes 1 60 bytes 36.802264065 0.000158 0													
	192.168.200.100 50906 192.168.200.150 979 2 134 bytes 552 1 74 bytes 1 60 bytes 36.831902219 0.000174 0													
	192.168.200.100 37560 192.168.200.150 978 2 134 bytes 125 1 74 bytes 1 60 bytes 36.790171438 0.000236 0													
	192.168.200.100 34346 192.168.200.150 977 2 134 bytes 228 1 74 bytes 1 60 bytes 36.800806664 0.000222 0													
	192.168.200.100 43870 192.168.200.150 976 2 134 bytes 471 1 74 bytes 1 60 bytes 36.824249404 0.000128 0													
	192.168.200.100 38366 192.168.200.150 975 2 134 bytes 134 1 74 bytes 1 60 bytes 36.790770268 0.000347 0													

Per ridurre gli impatti dell’attacco attuale e mitigare attacchi futuri, si può: bloccare l’IP attaccante sul firewall mettendo anche regole che limitano la frequenza di connessioni SYN; spegnere il servizio che manda in broadcast il nome del computer ricevente; installare un IDS/IPS con delle regole che riconoscono i pattern tipici di uno scan di porte.

Gianpaolo Scagione